1.Create a class Box that uses a parameterized constructor to initialize the dimensions of a box.The dimensions of the Box are width, height, depth. The class should have a method that can return the volume of the box. Create an object of the Box class and test the functionalities.

```java
package main;
class Box{
    int width;
    int height;
    int depth;
    Box(int width,int height,int depth){
        this.width = width;
        this.height = height;
        this.depth = depth;
    }
    public int volume() {
        return width*height*depth;
    }
}
public class Main {
    public static void main(String[] args) {
        Box obj = new Box(10,20,30);
        System.out.print("Volume : "+obj.volume());
    }
}
```

```
Volume : 6000
```

2.Create a new class called Calculator with the following methods: 1. A static method called powerInt(int num1,int num2) This method should return num1 to the power num2. 2. A static method called powerDouble(double num1,int num2). This method should return num1 to the power num2. 3. Invoke both the methods and test the functionalities. Hint: Use Math.pow(double,double) to calculate the power.

```java
package main;
class Calculator{
    public static int powerInt(int num1,int num2){
        return (int)Math.pow(num1,num2);
    }
    public static double powerDouble(double num1,int
            num2){
        return Math.pow(num1,num2);
    }
}
public class Main {
    public static void main(String[] args) {
        System.out.println(Calculator.powerInt(12, 2));
        System.out.println(Calculator.powerDouble(12.0, 2));
    }
}
```

```
144
144.0
```

3.Design a class that can be used by a health care professional to keep track of a patient's vital statistics. The following are the details. Name of the class - Patient Member Variables - patientName(String),height(double),width(double) Member Function - double computeBMI() The above method should compute the BMI and return the result. The formula for computation of BMI is weight (in kg) ÷ height*height(in metres). Create an object of the Patient class and check the results.

```java
package main;
class Patient {
    String patientName;
    double height;
    double weight;
    Patient(String patientName, double height, double weight){
        this.patientName = patientName;
        this.height = height;
        this.weight = weight;
    }
    public double computeBMI(){
        return weight / (height * height);
    }
}
public class Main {
    public static void main(String[] args) {
        Patient obj = new Patient("Balaji",1.75,70.0);
        double ans = obj.computeBMI();
        System.out.printf("%.3f", ans);
    }
}
```

```
<terminated>
22.857
```

4.Write a Program to check whether a given String is Palindrome or not.

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        String str = "madam";
        int l = 0;
        int r = str.length()-1;
        boolean flag = true;
        while(l<=r) {
            if(str.charAt(l)!=str.charAt(r)) {
                System.out.print("Not a Palindrome");
                flag = false;
                break;
            }
            l++;
            r--;
        }
        if(flag) {
            System.out.print("Palindrome");
        }
    }
}
```

```
Palindrome
```

5.Given a string, return a new string made of 'n' copies of the first 2 chars of the original string where 'n' is the length of the string.
Example1) i/p:Wipro o/p:WiWiWiWiWi

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        String str = "wipro";
        int n = str.length();
        String substr = str.substring(0,2);
        StringBuilder sb = new StringBuilder();
        for(int i=0;i<n;i++) {
            sb.append(substr);
        }
        System.out.print(sb.toString());
    }
}
```

```
wiwiwiwiwi
```

6.Write a java program that will return the first half of the string, if the length of the string is even. It should return null for odd length string. Example1) i/p:TomCat o/p:Tom Example2) i/p:Apron o/p:null

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        String str = "tomcat";
        int n = str.length();
        if(n%2==0) {
            System.out.print(str.substring(0,n/2));
        }
        else {
            System.out.print("null");
        }
    }
}
```

```
tom
```

7.Write a java program that accepts a string and returns a new string without the first and last character of the input string. Example1) i/p:Suman o/p:uma

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        String str = "suman";
        String ans = str.substring(1,str.length()-1);
        System.out.print(ans);
    }
}
```

```
uma
```

8.Given 2 strings, a and b, return a new string of the form short+long+short, with the shorter string on the outside and the longer string on the inside. The strings will not be the same length, but they may be empty (length 0).If input is "hi" and "hello", then output will be "hihellohi".

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        String a = "hi";
        String b = "hello";
        if(a.length()<b.length()) {
            System.out.print(a+b+a);
        }
        else {
            System.out.print(b+a+b);
        }
    }
}
```

```
hihellohi
```

9.Given a string, if the first or last chars are 'x', return the string without those 'x' chars, otherwise return the string unchanged. If the input is "xHix", then output is "Hi". If the input is "America", then the output is "America".

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        String str = "xhihellox";
        if(str.charAt(0)=='x' && str.charAt(str.length()-1)=='x') {
            System.out.print(str.substring(1,str.length()-1));
        }
        else {
            System.out.print(str);
        }
    }
}
```

```
hihello
```

10.Write a Java program that accepts a string (with * in it). The program should return a new string in which the following characters are removed *,the characters that are to the left and right of * Example1) i/p:ab*cd o/p:ad

```java
package main;
public class Main {
    public static void main(String[] args) {
        String str = "ab*cd";
        StringBuilder sb = new StringBuilder(str);
        int index;
        while((index=sb.indexOf("*"))!=-1) {
            sb.delete(index-1,index+2);
        }
        System.out.print(sb.toString());
    }
}
```

```
ad
```